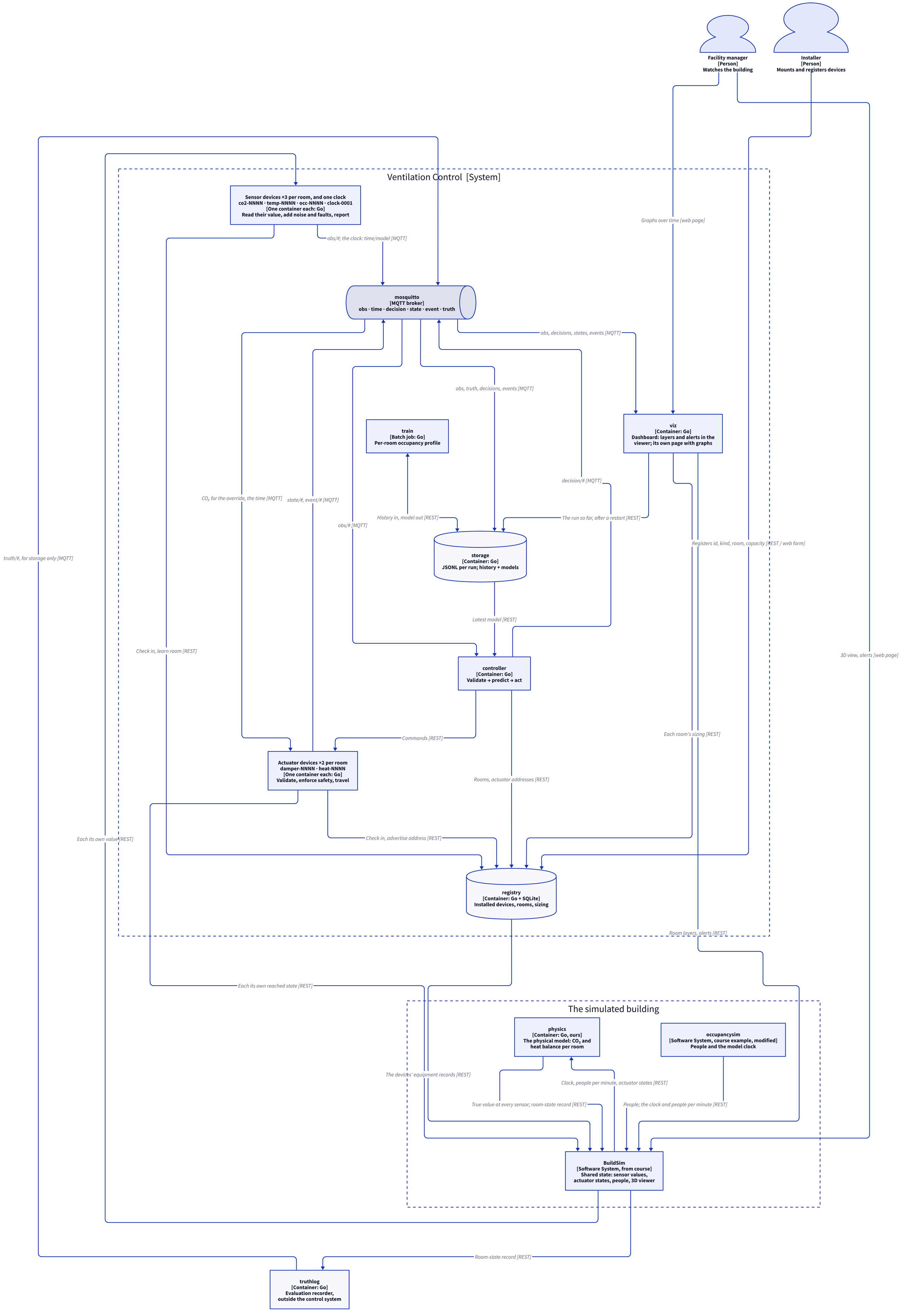

Facility manager
[Person]
Watches the building
Installer
[Person]
Mounts and registers devices
Ventilation Control [System]
Sensor devices ×3 per room, and one clock
co2-NNNN · temp-NNNN · occ-NNNN · clock-0001
[One container each: Go]
Read their value, add noise and faults, report
obs/#; the clock: time/model [MQTT]
mosquitto
[MQTT broker]
obs · time · decision · state · event · truth
obs, decisions, states, events [MQTT]
obs, truth, decisions, events [MQTT]
train
[Batch job: Go]
Per-room occupancy profile
viz
[Container: Go]
Dashboard: layers and alerts in the viewer; its own page with graphs
Graphs over time [web page]
decision/# [MQTT]
CO₂ for the override, the time [MQTT]
state/#, event/# [MQTT]
History in, model out [REST]
The run so far, after a restart [REST]
obs/# [MQTT]
Registers id, kind, room, capacity [REST / web form]
truth/#, for storage only [MQTT]
storage
[Container: Go]
JSONL per run; history + models
Latest model [REST]
Check in, learn room [REST]
3D view, alerts [web page]
controller
[Container: Go]
Validate → predict → act
Commands [REST]
Each room's sizing [REST]
Actuator devices ×2 per room
damper-NNNN · heat-NNNN
[One container each: Go]
Validate, enforce safety, travel
Rooms, actuator addresses [REST]
Each its own value [REST]
Check in, advertise address [REST]
registry
[Container: Go + SQLite]
Installed devices, rooms, sizing
Room layers, alerts [REST]
Each its own reached state [REST]
The simulated building
physics
[Container: Go, ours]
The physical model: CO₂ and heat balance per room
occupancysim
[Software System, course example, modified]
People and the model clock
The devices' equipment records [REST]
Clock, people per minute, actuator states [REST]
True value at every sensor; room-state record [REST]
People; the clock and people per minute [REST]
BuildSim
[Software System, from course]
Shared state: sensor values, actuator states, people, 3D viewer
Room-state record [REST]
truthlog
[Container: Go]
Evaluation recorder, outside the control system